ANIL KUMAR

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Professional Summary

Backend-focused Software Engineer with hands-on experience designing secure, scalable distributed systems and full-stack applications. Proven track record of eliminating critical security vulnerabilities (IDOR, PII exposure), engineering high-throughput data pipelines with Node.js, Fastify, and Firestore, and building production RAG systems with LangChain and vector databases. Strong foundation in data structures, algorithms (LeetCode Contest Rating: 1825, Top 7.16%), system design, and AWS (S3, DynamoDB, EC2), with demonstrated ability to independently own features from design through deployment.

Education

Indian Institute of Information Technology, Raichur 2023 – 2027
B.Tech – Artificial Intelligence & Data Science (Pursuing Engineering Degree)
Coursework: Data Structures & Algorithms, Object-Oriented Programming, Databases, Operating Systems, Computer Networks

Skills

- **Languages:** JavaScript, TypeScript, Python, C++, C
- **Backend & APIs:** Node.js, Express.js, Fastify, FastAPI, REST, Microservices
- **Frontend:** React.js, Next.js, TanStack Query, Tailwind CSS
- **Databases:** PostgreSQL, MySQL, MongoDB, Redis, Firestore, DynamoDB
- **Cloud & Infra:** AWS (EC2, S3, DynamoDB), Docker, Firebase Functions, Prometheus, CI/CD (Jenkins)
- **AI/ML & RAG:** LangChain, ChromaDB, Gemini API, Semantic Embeddings, RAG
- **Systems & Security:** RBAC, JWT Auth, Distributed Systems, Atomic Transactions, IDOR/PII Remediation
- **Tools:** Git, Jira, Socket.io, Postman

Experience

The Circle NextGen Pvt Ltd Jun 2026 – Present
Software Developer Intern

- Eliminated IDOR and PII-exposure vulnerabilities across support, analytics, and partner-profile APIs (Node.js, Fastify, Firestore) by enforcing ownership-based access checks, achieving 100% tenant data isolation with full unit-test coverage.
- Cut account-revocation latency from 5 days to real-time by removing an insecure JWT fast-path and enforcing live database validation in auth middleware, closing a critical session-hijacking risk.
- Prevented server OOM crashes at scale by designing a reusable cursor-based pagination utility for Firestore-backed Cloud Functions, lifting notification delivery from a hard 500-record cap to 100% of confirmed users.
- Eliminated phantom transaction failures by refactoring checkout logic into a single atomic Firestore transaction, achieving 100% consistency between order state and promo-code usage under retry conditions.
- Implemented end-to-end RBAC across client (React) and API (Fastify) layers, closing a direct-URL authorization bypass and achieving full route-guard coverage across 3 partner dashboards.
- Replaced a fully static sales dashboard with live data pipelines across 3 API endpoints, shipping a real-time Promoter Performance Comparison Matrix using React, TanStack Query, and Fastify.

Optifyx Technology Sep 2024 – Dec 2024
Full Stack Developer Intern

- Designed secure, high-performance RESTful APIs and optimized system workflows, reducing resolution time for customer operations by ∼30%.
- Built complex UI components and microservices, integrating real-time notification architectures using Git and Jira workflows.

Projects

Distributed Event Scheduling & Performance Engine github.com/Anilkumar7877/Temporal-Fault-Engine
Node.js, PostgreSQL, Redis, Prometheus, Docker

- Designed and developed a repeatable, high-performance distributed architecture enforcing strict reliability semantics across parallel nodes.
- Conducted extensive system monitoring, capacity planning, and automated debugging to process 10,000-event stress testing workloads while maintaining p99 execution latency below 110ms.

RAG Document Q&A System github.com/Anilkumar7877/RAG-QA-System
Python, FastAPI, LangChain, ChromaDB, Gemini API

- Implemented complex internal algorithms for semantic embeddings, building secure ingest pipelines capable of parsing large-scale enterprise data.
- Evaluated new technologies to continuously enhance model memory retention and achieve low-latency, page-attributed vector lookups.

Real-Time Voice AI Agent with Barge-In & RAG github.com/Anilkumar7877/voice-agent
Python, LangChain/LangGraph, Groq (Whisper + Llama 3.3), Cartesia TTS, ChromaDB, Silero VAD

- Engineered a real-time conversational voice pipeline (streaming STT → LLM → TTS) with sub-second barge-in interruption handling via Silero VAD and cooperative cancellation across synchronous APIs, enabling natural mid-sentence interruption instead of rigid turn-taking.
- Built an LLM-routed RAG and web-search tool-calling layer (LangGraph, ChromaDB, Tavily) with multi-turn conversational memory, achieving 100% tool-selection accuracy across a 12-case evaluation suite.
- Designed a git-commit-tagged evaluation harness (SQLite, LLM-as-judge scoring) that diagnosed a redundant model-reinitialization bug and validated a 58% reduction in average response latency (3.18s → 1.33s).

Achievements

- Core Algorithm Ingenuity: Solved 500+ DSA problems across LeetCode & GFG (LeetCode Contest Rating: 1825, Top 7.16%; GFG Score: 708).
- Team Leadership: Coding Club Lead – Managed 15+ students, spearheaded technical sessions on construction and debugging best practices, and drove participation up by 40%.

Position of Responsibility

Team Head, Entrepreneurship Cell Jul 2025 – Present

- Spearheaded cross-functional collaborations to drive organization-wide strategic initiatives across 15+ members.